

1(i)

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 8 \\ 1 \end{pmatrix} \quad l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \alpha \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$$

$$\overrightarrow{OB} = \begin{pmatrix} 6 \\ 8 \\ 4 \end{pmatrix} \quad \overrightarrow{OA} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \quad \overrightarrow{OC} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \alpha \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} \text{ for some } \alpha \in \mathbb{R}^+$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 5 \\ 8 \\ 1 \end{pmatrix} \quad \overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \alpha \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$$

$$|\overrightarrow{AB}| = \sqrt{5^2 + 8^2 + 1^2} = \sqrt{90} \quad |\overrightarrow{AC}| = \sqrt{(3\alpha)^2 + \alpha^2} = \sqrt{10\alpha^2}$$

$$\text{Since } |\overrightarrow{AB}| = |\overrightarrow{AC}|,$$

$$10\alpha^2 = 90$$

$$\alpha^2 = 9$$

At point C, $\alpha = 3$ (since $\alpha > 0$.)

$$\overrightarrow{OC} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + 3 \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 10 \\ 3 \\ 3 \end{pmatrix} \quad (\text{Shown})$$

(ii)

$$\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BD}$$

$$= \overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BD}$$

$$= \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \begin{pmatrix} 5 \\ 8 \\ 1 \end{pmatrix} + \begin{pmatrix} 9 \\ 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 15 \\ 11 \\ 4 \end{pmatrix}$$

OR

$$\overrightarrow{BD} = \overrightarrow{AC}$$

$$\overrightarrow{OD} = \overrightarrow{OC} - \overrightarrow{OA} + \overrightarrow{OB}$$

$$= \begin{pmatrix} 10 \\ 3 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \begin{pmatrix} 6 \\ 8 \\ 4 \end{pmatrix} = \begin{pmatrix} 15 \\ 11 \\ 4 \end{pmatrix}$$

(iii)

$$\sin \hat{BAC} = \frac{|\overrightarrow{AB} \times \overrightarrow{AC}|}{|\overrightarrow{AB}| |\overrightarrow{AC}|} = \frac{\left| \begin{pmatrix} 5 \\ 8 \\ 1 \end{pmatrix} \times \begin{pmatrix} 9 \\ 3 \\ 0 \end{pmatrix} \right|}{\sqrt{90} \sqrt{90}} = \frac{\left| \begin{pmatrix} -3 \\ 9 \\ -57 \end{pmatrix} \right|}{90} = \frac{\sqrt{(-3)^2 + 9^2 + (57)^2}}{90} = \frac{\sqrt{371}}{30}$$

$$\text{Area of parallelogram } ABDC = |\overrightarrow{AC} \times \overrightarrow{AB}|$$

$$= |\overrightarrow{AB}| |\overrightarrow{AC}| \sin \hat{BAC}$$

$$= \sqrt{90} \cdot \sqrt{90} \cdot \frac{\sqrt{371}}{30} = 3\sqrt{371}$$

2(a)	$A = \text{Total area of rectangles}$ $= \frac{1}{n} \left[\left(\frac{1}{n} \right)^2 e^{\frac{1}{n}} + \left(\frac{2}{n} \right)^2 e^{\frac{2}{n}} + \left(\frac{3}{n} \right)^2 e^{\frac{3}{n}} + \dots + \left(\frac{n-1}{n} \right)^2 e^{\frac{n-1}{n}} \right]$ $= \frac{1}{n^3} \left[1^2 e^{\frac{1}{n}} + 2^2 e^{\frac{2}{n}} + 3^2 e^{\frac{3}{n}} + \dots + (n-1)^2 e^{\frac{n-1}{n}} \right]$ $= \frac{1}{n^3} \sum_{r=1}^{n-1} \left(r^2 e^{\frac{r}{n}} \right)$ Limiting value of A $= \int_0^1 x^2 e^x \, dx$ $= \left[x^2 e^x \right]_0^1 - \int_0^1 2x e^x \, dx$ $= e - \left[\left[2x e^x \right]_0^1 - \int_0^1 2e^x \, dx \right]$ $= e - \left[2e - \left[2e^x \right]_0^1 \right]$ $= e - [2e - (2e - 2)]$ $= e - 2$ (b) $\text{Volume} = \pi \int_0^1 \left[e^2 - (x^2 e^x)^2 \right] dx$ $= 19.77$
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3(a)

$$\begin{aligned}
 \left(\frac{1-3x}{2+x}\right)^n &= (1-3x)^n (2+x)^{-n} \\
 &= \frac{1}{2^n} \left(1 - \frac{nx}{2} + \frac{n(n+1)}{8}x^2 + \dots\right) \left(1 - 3nx + \frac{9n(n-1)}{2}x^2 + \dots\right) \\
 &= \frac{1}{2^n} \left(1 - 3nx + \frac{9n(n-1)}{2}x^2 - \frac{nx}{2} + \frac{3n^2x^2}{2} + \frac{n(n+1)}{8}x^2\right) \\
 &= \frac{1}{2^{n+3}} (8 - 28nx + 49n^2x^2 - 35nx^2)
 \end{aligned}$$

$$\text{Validity: } \left|\frac{x}{2}\right| < 1 \Rightarrow |x| < 2 \Rightarrow -2 < x < 2$$

(b)(i)

$$\ln y = \frac{\tan^{-1} x}{2}$$

Diff implicitly,

$$\left(\frac{2}{y}\right) \frac{dy}{dx} = \frac{1}{1+x^2}$$

$$2(1+x^2) \frac{dy}{dx} = y$$

Diff implicitly,

$$2(1+x^2) \frac{d^2y}{dx^2} + 4x \frac{dy}{dx} = \frac{dy}{dx}$$

$$2(1+x^2) \frac{d^2y}{dx^2} = (1-4x) \frac{dy}{dx} \quad (\text{Shown})$$

(ii)

Diff implicitly,

$$2(1+x^2) \frac{d^3y}{dx^3} + 4x \frac{d^2y}{dx^2} = (1-4x) \frac{d^2y}{dx^2} - 4 \frac{dy}{dx}$$

$$\text{When } x=0, y=1, \frac{dy}{dx} = 1/2, \frac{d^2y}{dx^2} = 1/4, \frac{d^3y}{dx^3} = -\frac{7}{8}.$$

$$\therefore y = 1 + \frac{x}{2} + \frac{x^2}{8} - \frac{7x^3}{48} + \dots$$

(iii)

$$\begin{aligned}
 e^{\left(\frac{\tan^{-1} x}{2}\right) - \ln(\sec x)} &= e^{\left(\frac{\tan^{-1} x}{2}\right)} e^{\ln(\sec x)^{-1}} = e^{\left(\frac{\tan^{-1} x}{2}\right)} (\cos x) \\
 &= \left(1 + \frac{x}{2} + \frac{x^2}{8} + \dots\right) \left(1 - \frac{x^2}{2} + \dots\right) \\
 &= \left(1 + \frac{x}{2} - \frac{3x^2}{8}\right)
 \end{aligned}$$

4(i)

$$y = \frac{2x^2 + kx - 1}{x + 1}$$

$$\frac{dy}{dx} = \frac{(x+1)(4x+k) - (2x^2 + kx - 1)(1)}{(x+1)^2} = \frac{2x^2 + 4x + k + 1}{(x+1)^2}$$

$$\text{Let } \frac{dy}{dx} = 0 \Rightarrow 2x^2 + 4x + k + 1 = 0$$

Now, discriminant $4^2 - 4(2)(k+1) = 8(1-k) < 0$ if $k > 1$
 $\therefore C$ has no stationary point if $k > 1$ [shown]

(ii)

$$y = \frac{2x^2 + kx - 1}{x + 1} = 2x + (k-2) + \frac{-k+1}{x+1}$$

Given $y = 2x + 1$ is an asymptote, $k - 2 = 1 \Rightarrow k = 3$ [shown]

(iii)

$$y = \frac{2x^2 + 3x - 1}{x + 1} = 2x + 1 - \frac{2}{x+1}$$

$$y = x - \frac{1}{x} \xrightarrow{A} y = 2\left(x - \frac{1}{x}\right) \xrightarrow{B} y = 2(x+1) - \frac{2}{(x+1)} \xrightarrow{C} y = \left(2x + 2 - \frac{2}{x+1}\right) - 1$$

$$= 2x - \frac{2}{x} \qquad = 2x + 2 - \frac{2}{x+1} \qquad = 2x + 1 - \frac{2}{x+1}$$

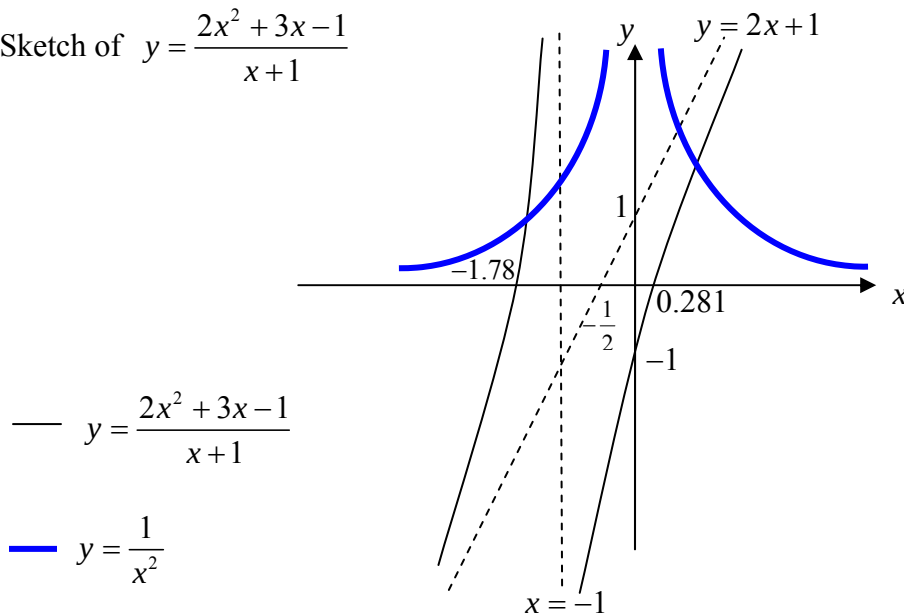
A: a stretch parallel to the y -axis by a scale factor 2.

B: a translation of -1 unit in the direction of x -axis.

C: a translation of -1 unit in the direction of y -axis.

(iv)

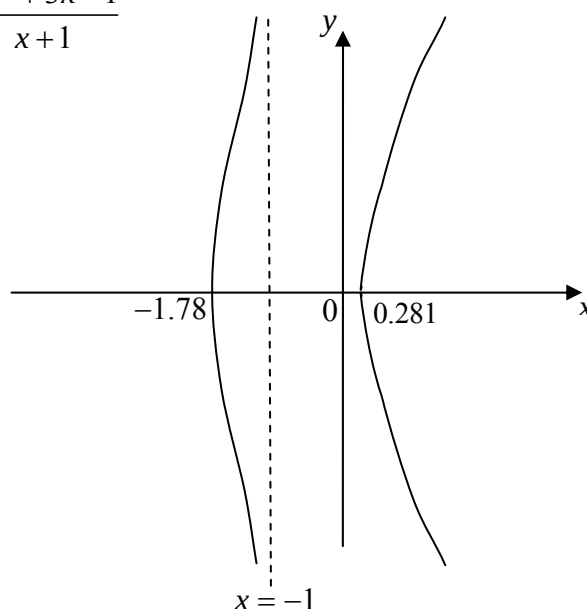
Sketch of $y = \frac{2x^2 + 3x - 1}{x + 1}$



— $y = \frac{2x^2 + 3x - 1}{x + 1}$

— $y = \frac{1}{x^2}$

Sketch of $y^2 = \frac{2x^2 + 3x - 1}{x + 1}$



$$2x^4 + 3x^3 - x^2 - x = 1 \Rightarrow \frac{2x^2 + 3x - 1}{x + 1} = \frac{1}{x^2}$$

Insert $y = \frac{1}{x^2}$.

There are 2 points of intersection, so the number of real roots = 2.

5(i) Non-random or Sample may be biased as the interviewer may simply select those employees who are easiest to interview or any other appropriate answers.

(ii) Use stratified sampling.
Draw random samples from each group as follows :

	No. of employees
Production	$\frac{1200}{2000} \times 100 = 60$
Marketing	$\frac{600}{2000} \times 100 = 30$
Management	$\frac{100}{2000} \times 100 = 5$
Others	$\frac{100}{2000} \times 100 = 5$

6(i)	Let μ be the population mean reading speed of Primary 2 school children. Sample mean, $\bar{w} = \frac{\sum w}{n} = \frac{13560}{80} = 169.5$ Unbiased estimate of population variance s^2 $= \frac{1}{n-1} \left[\sum w^2 - \frac{(\sum w)^2}{n} \right] = \frac{1}{79} \left[2388670 - \frac{(13560)^2}{80} \right] = \frac{90255}{79} = 1142.4$ $H_0 : \mu = 163$ $H_1 : \mu > 163$ Under H_0, by CLT, $\bar{W} \approx N(\mu, \frac{\sigma^2}{n}) \approx N\left(\mu, \frac{s^2}{n}\right)$ and test statistic $Z = \frac{\bar{W} - \mu}{\frac{s}{\sqrt{n}}} \sim N(0, 1)$ where $\mu = 163$, $\bar{w} = 169.5$, $s = \sqrt{1142.4}$, $n = 80$. At 5% level of significance, we use a right-tailed z-test and reject H_0 if p-value < 0.05. From GC, p-value = 0.0427 Since p – value = 0.0427 < 0.05, we reject H_0 and conclude that at 5% level of significance there is sufficient evidence to suggest that the mean reading speed of Primary 2 school children has increased. (ii) ‘At the 5% level of significance’ means that there is a probability of 0.05 to wrongly conclude that the mean reading speed has increased when in fact it has not. (iii) The test will not be valid since the sample drawn is not a good representation of the population of Primary 2 school children / the standard of children is different for different schools or any other logical reason.
7(a) (i) (ii)	No of ways = ${}^{10}C_1 \times {}^9C_3 = 840$ OR ${}^{10}C_3 \times {}^9C_1 = 840$ Form CLC unit: ${}^3C_2 \times 2$ ways Seat 8 units in a round table: $(8-1)!$ ways Total number of ways = ${}^3C_2 \times 2 \times (8-1)! = 30240$ Total number of ways = $30240 \times 10 = 302400$

7(b)	Case 1: 4 digit number, ending with 2: No of ways = $2 \times 3 \times 2 \times 1 = 12$ Case 2: 4 digit number, does not end with 2: No of ways = $2 \times (2 + 1) \times 1 = 6$ Case 3: 5 digit number, ending with 2: No of ways = $4! \times 1 = 24$ Case 4: 5 digit number, does not end with 2: No of ways = $\frac{4!}{2!} \times 2 = 24$ Total number of ways = $12 + 6 + 24 + 24$ $= 66$
8(a) (i) (ii) (iii) (b) (c)	$P(A) = \frac{9}{36} = \frac{1}{4}$ $P(B) = \frac{4}{36} = \frac{1}{9}$ $P(A \cap B) = \frac{1}{36}$ Since $P(A \cap B) = P(A) P(B)$, they are independent. $A \cup B$ represents the event the card taken is either blue or numbered 1. $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{9}{36} + \frac{4}{36} - \frac{1}{36} = \frac{1}{3}$ $P(A' B') = \frac{P(A' \cap B')}{P(B')} = \frac{1 - \frac{1}{3}}{\frac{32}{36}} = \frac{3}{4}$ <u>Method 1</u>: Required probability = $1 \times \frac{3}{35} \times \frac{2}{34} = \frac{3}{595}$ or 0.00504 <u>Method 2</u>: Required probability = $\frac{4}{36} \times \frac{3}{35} \times \frac{2}{34} \times 9 = \frac{3}{595}$ <u>Method 3</u>: Required probability = $\binom{4}{3} \left(\frac{1}{36}\right) \left(\frac{1}{35}\right) \left(\frac{1}{34}\right) 3! \times 9 = \frac{3}{595}$ <u>Method 4</u>: Required probability = $\frac{9 \binom{4}{3}}{\binom{36}{3}} = \frac{3}{595}$ <u>Method 1</u>: Required probability = $\frac{{}^{32}C_{20}}{{}^{36}C_{20}} = \frac{52}{1683} = 0.030897 = 0.0309$ <u>Method 2</u>: Required probability $= \frac{32}{36} \times \frac{31}{35} \times \frac{30}{34} \times \frac{29}{33} \times \frac{28}{32} \times \dots \times \frac{17}{21} \times \frac{16}{20} \times \frac{15}{19} \times \frac{14}{18} \times \frac{13}{17}$ $= \frac{16}{36} \times \frac{15}{35} \times \frac{14}{34} \times \frac{13}{33} = 0.030897 = 0.0309$

9(a) (b) (c)	Let X be the no. of customers who drink coffee. Then for 7 customers, $X \sim B(7, \frac{1}{4})$ $P(X \geq 3) = 1 - P(X \leq 2) = 0.244$ (3sf) For 500 customers, $X \sim B(500, \frac{1}{4})$ Since $n = 500$ is large, $np > 5$ and $nq > 5$, $X \sim N(500 \times \frac{1}{4}, 500 \times \frac{1}{4} \times \frac{3}{4}) = N(125, 93.75)$ approximately. Now, given $P(X < k) \geq 0.99$, Using continuity correction, $P(X < k - 0.5) \geq 0.99$ By GC, $k - 0.5 \geq 147.524 \Rightarrow k \geq 148.024$ The least value of $k = 149$ Let W be the no. of customers who make a complaint. For 500 customers, $W \sim B(500, \frac{1}{1000})$ Since n is large and $np = \frac{1}{2} < 5$, $W \sim \text{Po}(\frac{1}{2})$ approximately. By GC, $P(W < 2) = P(W \leq 1) = 0.910$ (3sf)
10 (a) (b)	Let X = Time in minutes Eric takes to solve a “Beginner” puzzle, Y = Time in minutes Eric takes to solve an “Intermediate” puzzle, W = Time in minutes Eric takes to solve an “Expert” puzzle. Then $X \sim N(3.2, 0.5^2)$, $Y \sim N(\mu, 2.1^2)$, $W \sim N(12.3, \sigma^2)$. $X + Y \sim N(3.2 + \mu, 0.5^2 + 2.1^2) = N(3.2 + \mu, 4.66)$ Given $P(X + Y > 9.5) = 0.8$ $\Rightarrow P(Z \leq \frac{9.5 - (3.2 + \mu)}{\sqrt{4.66}}) = 0.2$ From GC, $\frac{6.3 - \mu}{\sqrt{4.66}} = -0.8416212335$ $\mu = 8.12 = 8.1$ (1 dp) Given $P(W < 10) \leq 0.25$ $\Rightarrow P(Z < \frac{10 - 12.3}{\sigma}) \leq 0.25$ From GC, $\frac{10 - 12.3}{\sigma} \leq -0.6744897495$ $\sigma \leq 3.41$ Maximum $\sigma = 3.4$ (1 dp)

(c)

Given $\mu = 7.8$ and $\sigma = 3.2$.

$$Y + W - 4X \sim N(7.8 + 12.3 - 4(3.2), 2.1^2 + 3.2^2 + 16(0.5^2)) = N(7.3, 18.65)$$

From GC, $P(Y + W > 4X) = P(Y + W - 4X > 0) = 0.955$.

(d)

$$\text{Average time, } \bar{T} = \frac{X_1 + X_2 + X_3 + Y_1 + Y_2 + W}{6}$$

$$E(\bar{T}) = \frac{3(3.2) + 2(7.8) + 12.3^2}{6} = 6.25$$

$$\text{Var}(\bar{T}) = \frac{3(0.5^2) + 2(2.1^2) + 3.2^2}{36} = \frac{19.81}{36}$$

$$\therefore \bar{T} \sim N(6.25, \frac{19.81}{36})$$

From GC, $P(\bar{T} \leq 7) = 0.844$

11(i)

$$PV^c = k$$

$$\ln P + c \ln V = \ln k$$

$$\ln V = \frac{\ln k}{c} - \frac{1}{c} \ln P$$

 $\therefore y = a + bx$ is a straight line, where $a = \frac{\ln k}{c}$ and $b = -\frac{1}{c}$ are constants.

[shown]

(ii)

Using G.C., enter values of P and V in list L1 and L2 .Let $L3 = \ln P$ and let $L4 = \ln V$

L1	L2	L3	1	L2	L3	L4	4
7.7	7.7	0		7.7	0	2.04122	
5.8	5.8	.69315		5.8	.69315	1.7579	
4.5	4.5	1.0986		4.5	1.0986	1.5041	
3.4	3.4	1.3863		3.4	1.3863	1.2238	
2.3	2.3	1.9459		2.3	1.9459	.83291	
1.9	1.9	2.3026		1.9	2.3026	.64185	
1.4	1.4	2.6391		1.4	2.6391	.33647	
L1(1)=1				L4(1)=2.041220328...			

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LinReg
y=a+bx
a=2.143358635
b=-.6621900222
r^2=.9877756941
r=-.9938690528

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$$r = -0.994$$

(iii) By using G.C., the required estimated regression line of y on x is

$$y = 2.1434 - 0.66219x$$

$$y = 2.14 - 0.662x \text{ (to 3 s.f.)}$$

$$\ln V = 2.1434 - 0.66219 \ln P$$

$$\ln V = 2.1434 - 0.66219 \ln(8) = 0.76641$$

$$V = e^{0.76641} = 2.15203 = 2.2 \text{ (to 1 d.p.) (ans)}$$

Since $r = -0.994 \approx -1$ which indicates that the sample values of x and y are almost perfectly linearly correlated and $\ln(1) \leq \ln(8) \leq \ln(14)$, therefore the prediction is reliable.

(iv)
$$\ln V = \frac{\ln k}{c} - \frac{1}{c} \ln P$$

$$y = 2.1434 - 0.66219x$$

$$\frac{1}{c} = 0.66219$$

$$c = \frac{1}{0.66219} = 1.51014 = 1.51 \text{ (to 3 s.f.)}$$

$$\frac{\ln k}{c} = 2.1434$$

$$\ln k = 2.1434 \times 1.51014 = 3.2368$$

$$k = e^{3.2368} = 25.452 = 25.5 \text{ (to 3 s.f.)}$$

(v) For $\sum (y - Y')^2$ to be minimum,

Then $Y' = a + bx$ must be $y = 2.1434 - 0.66219x$

By using G.C., let $L5 = 2.1434 - 0.66219 L3$ and $L6 = (L4 - L5)^2$

L4	L5	L6	Σ(L6)
2.0412	2.1434	.001044	.028156283
1.7579	1.6844	.0054	
1.5041	1.4159	.00777	
1.2238	1.2254	2.7E-6	
.83291	.85484	4.8E-4	
.64185	.61865	5.4E-4	
.33647	.39584	.00352	
L6(1) = .0104406851...			

Minimum value of $\sum (y - Y')^2 = 0.0282$ (to 3 s.f.)